

Comparative Study on Agility, Efficiency, and Impact Absorption of Bipedal Robots with Active Toes

Jung-Gil Kim, Wontae Ye, Geunwoo Cho, Seong-Ho Yun, Se-Hyoung Cho, and Yong-Jae Kim

Abstract— Human legs exhibit high efficiency, agility, and impact absorption, with toes playing a crucial role in these capabilities. While many attempts have been made to implement human-like toes in robots, they have not fully replicated human characteristics nor rigorously validated their benefits. In this study, we propose a 14-degree-of-freedom biped robot that emulates the lightweight, high-torque, and robust nature of human toes. To quantitatively analyze the effectiveness of the proposed structure in terms of agility, efficiency, and impact absorption, we carefully developed a simulation model and reward functions that minimize the sim-to-real gap to reflect the actual actuators with coupled transmissions, feasible contact with ground, and accurate power consumption. The reinforcement learning (RL)-based training framework was designed to minimize subjective developer biases, ensuring objective performance evaluation through a streamlined, goal-oriented reward function. The training process was implemented in the Isaac Lab environment, with policies optimized using the Proximal Policy Optimization (PPO) algorithm. The simulation results indicate that the bipedal robot with active toes demonstrated 11.3 % and 33.8 % reductions in Cost of Transport (CoT) and maximum GRF (ground reaction force) at 1.3m/s walking and a 10.2 % reduction in completion time for agility motions compared to the same robot without toe joints.

I. INTRODUCTION

Human locomotion is an intricate and highly efficient process, with the foot playing a crucial role in agility, efficiency, and impact absorption [1][2]. The human foot weighs approximately 1 kg and is composed of a 2 degree-of-freedom (DOF) ankle and multiple DoF toes [1]. Despite its relatively light weight, the foot can deliver up to 104 Nm of isometric torque and withstand ground reaction forces equivalent to 2.5 times body weight during running [1]-[3]. The human toe, in particular, contributes significantly to stride length, stability, and propulsion, allowing for dynamic adaptation to uneven terrain [4]. Inspired by human's capabilities, bipedal robots have been actively developed in recent years, demonstrating increasingly natural and complex walking behaviors due to advancements in machine learning-based locomotion controllers [5]–[11].

Despite these advancements, most bipedal robots employ simple foot structures considering mechanical simplicity and robustness. For instance, Cassie has a single-DoF ankle [10], and Digit features a two-DoF ankle [12], both actively

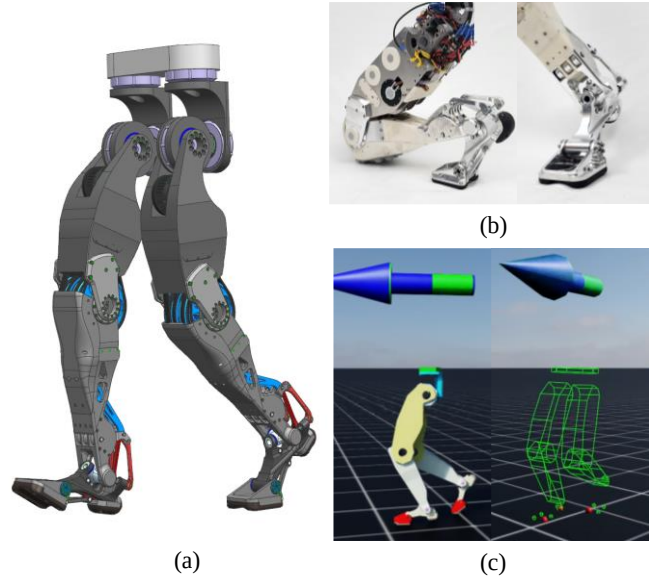


Fig. 1. (a) Detailed mechanical design of the proposed 14-DOF biped robot. (b) HyperLeg mechanism with a 2-DOF ankle and a 1-DOF toe, (c) The simulation model rigorously matched with actual robot system.

controlled for balance. However, neither incorporates an active toe joint, limiting foot-ground adaptability. As a result, bipedal robots with active toe DOFs remain largely unexplored. Research on robotic toes has been ongoing for decades, with studies demonstrating their benefits in increasing stride length, shock absorption and improving gait naturalness [13]–[16]. Active toe mechanisms have been explored, but they increase complexity, distal mass and structural fragility, making them challenging for practical implementation in high-performance robots [17]. To address these limitations, passive and spring-loaded toe mechanisms have been introduced to improve locomotion efficiency and terrain adaptability with relatively simple and light weight design [18][19]. However, these approaches lack the dynamic force modulation and active control seen in human toes.

To address these limitations, we previously developed HyperLeg, a leg mechanism featuring two DOF in the ankle and one DOF in the toe, as shown in Fig. 1(b) [20]. This design introduced a novel toe actuation system that mimics the high torque and lightweight structure of the human foot, providing enhanced impact resilience and agility. Building

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upon this foundation, this study presents a fully articulated 14-DoF bipedal robot, as depicted in Fig. 1(a). The proposed leg design includes a custom high-torque actuator for the hip joint, developed to optimize dynamic movement and power efficiency. The total leg length measures 786 mm, with each leg weighing 8.27 kg. When combined with a 15.46 kg upper body, the robot's total mass is 32 kg, ensuring a realistic weight distribution for human-scale bipedal locomotion.

An adult weighing approximately 70 kg, walking at a speed of 1.33 m/s, consumes 125 W of power [1]. The energy expenditure during locomotion can be quantified by the Cost of Transport (CoT). Humans exhibit highly efficient walking with a CoT of 0.316, which is significantly lower than that of traditional bipedal robots, whose CoT values are typically much higher than 1 [21][22]. To quantitatively analyze the effectiveness of the proposed structure in terms of agility, efficiency, and impact absorption, we set up a high-fidelity simulation model that minimizes the sim-to-real gap to accurately reflect the physical hardware and actuators. The model enables comparison between bipedal robots with and without active toe joints, and the reinforcement learning (RL)-based training framework was designed to minimize subjective developer biases, ensuring objective performance evaluation through a streamlined, goal-oriented reward function.

The remainder of this paper is organized as follows: Section II presents the detailed design of the proposed bipedal robot and provides details on the developed actuator. Section III describes the simulation formulation and Section IV reports the simulation results with an in-depth analysis. Finally, Section V concludes the paper.

II. HARDWARE

A. Mechanical Design

As shown in Fig. 2(a) and (b), the bipedal robot consists of a hip with three degrees of freedom (DOF), a knee with one DOF, an ankle with two DOFs, and a toe with one DOF. The leg length up to the hip pitch joint is 786 mm, and the total height is 958.6 mm. Each leg weighs 8.27 kg, and since all heavy actuators are housed within the thigh frames, the leg features lightweight distal parts, similar to those of humans. As can be seen in Fig. 2(b), despite having an active toe, the distal part remains lightweight, similar to that of a human, making it well-suited for agile and natural walking. Power is transmitted through belts and linkages, and as shown in Fig. 2(c), the mechanism allows for a wide range of motion. Figs. 2(d), (e), and (f) illustrate the structure of the actuators and power transmission mechanisms driving the knee, ankle, and toe, respectively. Since the ankle has two DOFs, the robot has the two identical structures shown in Fig. 2(e) in parallel, enabling differential actuation for ankle roll and pitch motions. For more detailed explanations, refer to the authors' previous work [20].

Fig. 3 illustrates the structure and actual images of the three types of actuators developed. As shown in Fig. 3(a), the actuator for the hip requires high torque output and is therefore implemented using a 25:1 cycloid gear and an

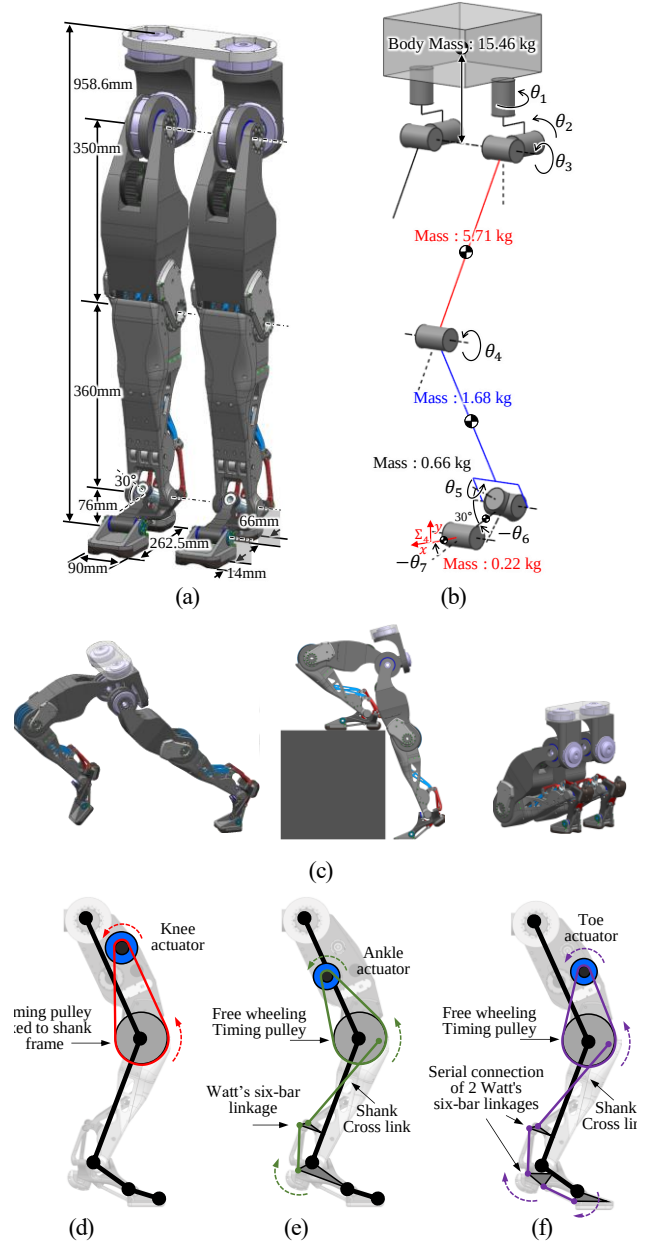


Fig. 2. (a) Size and shape of the 14-DOF biped robot utilizing HyperLeg mechanism. (b) Coordinates and mass distributions. (c) Human-like wide range of motions. (d), (e), and (f) Coupled belt and linkage mechanisms for the knee, ankle and toe actuations, respectively [20].

outrunner frameless motor with a large air gap diameter. The structures of the actuators for the knee and those for the ankle and toe are shown in Figs. 3(b) and (c), respectively. Both are composed of a 7:1 planetary gear and an inrunner BLDC motor with low rotor inertia, achieving additional reduction ratios through the power transmission mechanism. All components were carefully selected and designed with considerations for low friction resistance, high backdrivability, and high-impact resistance while maintaining high power output. These features play a crucial role in minimizing the sim-to-real gap. The detailed specifications of the robot and actuators are summarized in Table I.

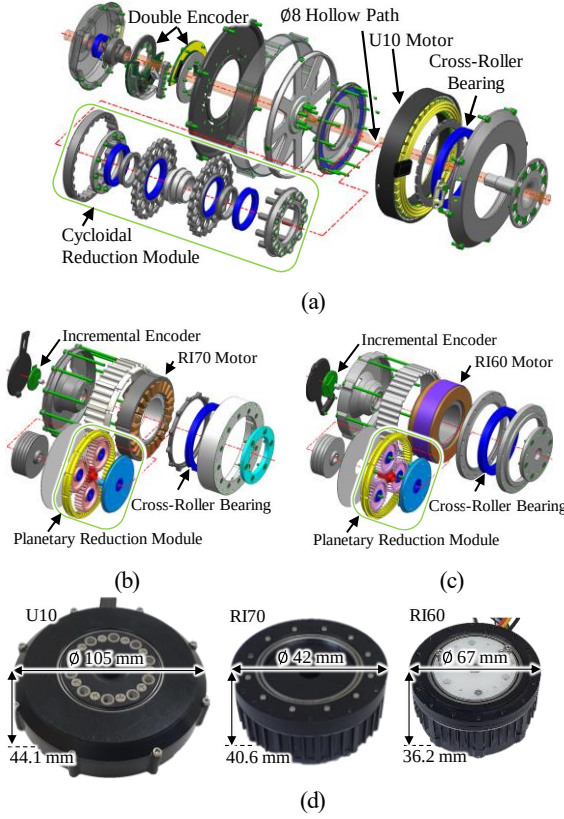


Fig. 3. (a) Actuator for hip yaw, roll, and pitch joints composed of out-runner motors and cycloid gears. (b) and (c) Actuators for the knee joint and the ankle and toe joints composed of in-runner motors and planetary gears, respectively. (d) Implemented Actuators.

B. Cooperative Actuation

As can be seen from the power transmission structures in Figs. 2(d), (e), and (f), all actuators for the knee, ankle, and toe are located in the thigh, reducing the weight of the distal segments and enhancing impact resistance against ground contact forces. Moreover, the actuators for the knee, ankle, and toe are intentionally coupled so that when the foot makes contact with the ground, all these actuators cooperate to extend the knee and perform ankle plantar flexion, generating a substantial propulsive force.

The Jacobian J_m , which represents the velocity relationship between the motors and joints, is expressed as follows:

$$\dot{\theta} = J_m \dot{\phi}, \quad (1)$$

where ϕ and θ represent the angles of the motors and joints, respectively. ϕ is a 7×1 vector that denotes the angles of the hip yaw, roll, and pitch motors, followed by the knee motor, left and right ankle motors, and the toe motor. Similarly, θ is a 7×1 vector representing the angles of the hip yaw, roll, and pitch joints, followed by the knee joint, ankle pitch and roll joints, and the toe joint.

The relationship between the motor torques τ_m and joint torques τ_{jt} can be expressed as:

$$\tau_m = J_m^T \tau_{jt} \quad (2)$$

An example of the motor torque and joint torque relationship

in a walking posture is given by the following equation:

$$\tau_{jt} = \begin{bmatrix} 25 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 25 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 25 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -31.11 & -31.11 & -31.11 & -31.11 \\ 0 & 0 & 0 & 0 & 26.69 & 26.69 & 19.23 \\ 0 & 0 & 0 & 0 & 16.51 & -16.51 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 21.27 \end{bmatrix} \tau_m \quad (3)$$

Since the actuators for the hip are connected in series, the upper-left 3×3 portion of the matrix contains only diagonal terms. However, the fourth row, which corresponds to knee actuation, is coupled with all the knee, ankle, and toe actuators, each contributing to knee motion through a reduction ratio of 31.11:1. Similarly, the cooperative output of the ankle and toe motors is reflected in the ankle pitch motion, as seen in the fifth row of the matrix. This intentional coupling structure is referred to as Cooperative Actuation (CA) [23].

III. LEARNING-BASED CONTROL

A. Simulation Setup Minimizing Sim-To-Real Gap

Although model-free learning-based control has surpassed traditional model-based control in many aspects, the presence of inaccurate robot dynamics and actuator models in simulation often leads to unrealistic learning outcomes or significant sim-to-real gaps upon deployment. In particular, parallel linkage transmission ankle structures [12] and CA structures like HyperLeg are frequently used in bipedal robots, but common machine learning simulation environments typically model robots as serial-link mechanisms with direct joint actuation, leading to errors in torque and speed estimation. Furthermore, simplified contact models often result in unrealistic locomotion policies that rely on simple ground reaction forces without ensuring stable surface contact, which is essential in real-world scenarios. Moreover, reward functions designed to encourage energy-efficient locomotion frequently simplify energy consumption by summing squared torques instead of utilizing an accurate CoT. Relying solely on domain randomization to bridge this sim-to-real gap results in overly conservative policies. Conversely, designing hardware that minimizes the sim-to-real gap - such as strictly serial-link robots - eliminates opportunities to leverage hardware advantages, limiting potential performance gains. To address these challenges, this paper focused on:

- Realistic actuator modeling that accurately reflects the relationship between joints and actuators,
- Integration of accurate CoT estimation into training, and
- Reward function design that encourages physically plausible contact conditions during locomotion.

By incorporating these improvements, we aim to reduce the sim-to-real gap while retaining the benefits of complex hardware architectures, ultimately enabling robust and energy-efficient real-world bipedal locomotion.

As shown by the blue rectangular region in Fig. 4(a), the motor output can be simply constrained by stall torque and no-load speed on the torque-speed graph. However, this restriction is unrealistic because it implies that the maximum

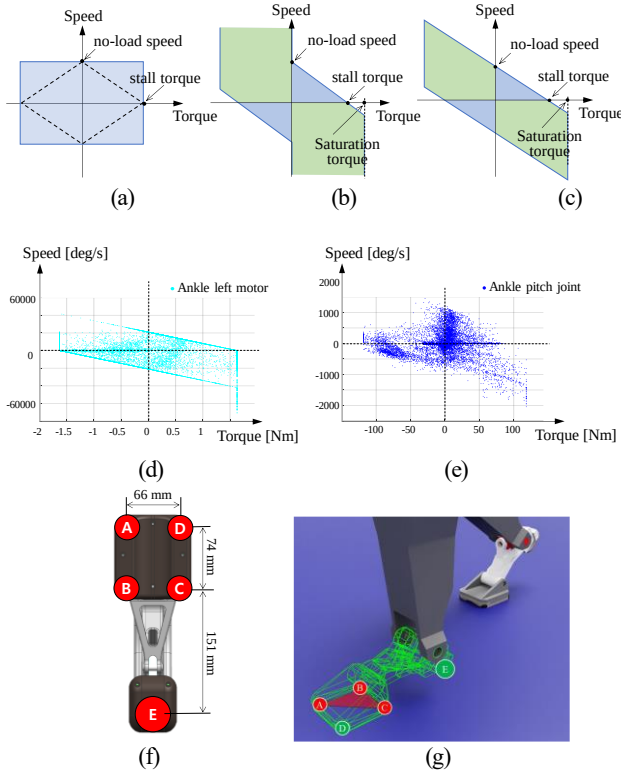


Fig. 4. (a), (b), and (c) speed-torque graphs used to impose motor torque output limits. (d) and (e) speed-torque graphs of the ankle motor and the ankle pitch joint for running at 5 m/s. (f) and (g) 5 contact points for the toe and heel.

torque and maximum speed can be achieved simultaneously. A more realistic constraint, considering the characteristics of a typical electro-magnetic motor, is represented by the dotted rhombus shape. However, this method imposes overly conservative constraints in the energy absorption region.

Various simulators offer different motor models. Fig. 4(b) shows the default motor output constraint model used in Isaac Lab, where it can be observed that the energy absorption region (marked in green) is unconstrained. In this study, we designed a realistic motor modeling approach that accurately reflects the characteristics of the BLDC motor and the custom motor controller we developed. As shown in Fig. 4(c), this method applies realistic constraints across all operating regions. Specifically, when the motor speed is ω and the target torque is τ , the actual motor output torque τ_{clip} is defined as follows:

$$\tau_{clip} = \max(\min(\tau, \tau_{max}), \tau_{min}), \quad (4)$$

$$\text{with } \tau_{min} = \min\left(\max\left(-\tau_{sat}, \tau_{stall}\left(-1 - \frac{\omega}{\omega_o}\right)\right), \tau_{sat}\right)$$

$$\tau_{max} = \max\left(\min\left(\tau_{sat}, \tau_{stall}\left(1 - \frac{\omega}{\omega_o}\right)\right), -\tau_{sat}\right)$$

where τ_{stall} refers to the stall torque, i.e., the maximum torque the motor can exert when it is stationary. τ_{sat} represents the saturation torque, which is the maximum torque the motor can produce before reaching magnetic saturation. ω and ω_o refer to the current motor speed and no-load speed respectively.

Fig. 4(d) illustrates an example of ankle motor output

during a simulation where the robot runs at 5 m/s. An important point to note is that, due to the effects of the CA structure, the motor torque does not have a one-to-one relationship with the joint torque. Therefore, in this study, the joint velocity from the simulator output and the target joint torque from the output of the network are converted to motor speed and torque using (1) and (3), and then the motor torque is clipped according to (4). Finally, the motor torque is reconverted to target joint torque for use in control as shown in Fig. 4(e). This method enables accurate output limitation for the CA structure without additional simulation computation overhead. Additionally, since HyperLeg exhibits dominant friction effects in joint space, the joint command torque used in simulation is computed as:

$$\tau_{cmd} = \tau_{j_clip} - \left(f_c \cdot \tanh\left(\frac{\dot{q}}{v_m}\right) + f_v \dot{q}\right) \quad (5)$$

where τ_{j_clip} represents the joint torque computed using (4) and (2), f_c denotes the Coulomb friction in the joint, f_v is the viscous friction in the joint, $\dot{q}$ is the joint speed, and τ_{cmd} is the friction-compensated joint torque. By applying this friction model, realistic torque dynamics can be accurately simulated, further enhancing the validity of learned locomotion strategies in real-world deployment.

For the policy to be effective, the resulting motion must be both energy-efficient and stable in the real world—an essential prerequisite for successful sim-to-real transfer. To estimate CoT precisely, the energy consumption during the robot's locomotion was analyzed along two distinct pathways: Joule heating and mechanical loss as following equation:

$$CoT = \frac{P}{mgv} = \frac{P_{joule} + P_{mech}}{mgv} \quad (6)$$

where P represents the amount of power consumption for locomotion, and m , g , and v denote the robot mass, gravitational acceleration, and robot speed, respectively. Since the denominator is the product of force and velocity, the CoT equation becomes dimensionless. The power consumption P is composed of the loss by Joule heating P_{joule} and the loss during mechanical work P_{mech} . Nonzero P_{mech} means that positive work is not fully retrieved by negative work by interaction with the ground. First, Joule heating generated by the current used to produce torque in the motor is calculated as

$$P_{joule} = I^2 R = (\tau/K_t)^2 R \equiv C_p \tau^2 \quad (7)$$

where I , R , and K_t represent the motor current, the winding resistance, and torque constant, respectively. $C_p = R/K_t^2$ is the power coefficient we defined. (6) is similar to the torque penalty commonly used in traditional RL, which is based on the 2-norm of torque. However, it differs in that the torque is converted into motor torque and scaled by the power coefficient, allowing for an accurate calculation of Joule heating loss. Second, the mechanically consumed power is given by the product of torque and angular velocity of motors:

$$P_{mech} = \begin{cases} \eta_{fwd}^{-1} \cdot \tau \cdot \omega & \text{if } \tau \cdot \omega \geq 0, \\ \eta_{inv} \tau \cdot \omega & \text{if } \tau \cdot \omega < 0, \end{cases} \quad (8)$$

where, η_{fwd} and η_{inv} represent the output efficiency and

Table I. Mechanical Specifications

Weight [kg]	Total		32 (one leg: 8.27, pelvis: 15.46)				
	pelvis		Thigh	Shank	Foot	Toe	
	15.46		5.71	1.68	0.66	0.22	
Joints	Hip yaw	Hip roll	Hip pitch	Knee	Ankle pitch	Ankle roll	Toe
Range of motion [deg]	-35~15	-5~90	-120~120	0~150	-56~60	-30~30	-65~10
Equivalent reduction Ratio	25.00 : 1	25.00 : 1	25.00 : 1	31.11 : 1	26.00 : 1*	16.51 : 1*	31.93 : 1*
No-load speed [deg/s]	17,189	17,189	17,189	18,750	20,970	20,970	20,970
Max torque with CA [Nm]	126.0	126.0	126.0	212.0	111.7	48.4	46.8
Coulomb friction [Nm]	3.30	3.30	3.30	5.59	2.09	2.2	0.88
Viscous friction [Nm/(rad/s)]	0.08	0.08	0.08	0.1	0.1	0.1	0.1
Motors	Hip yaw	Hip roll	Hip pitch	Knee	Ankle left	Ankle right	Toe
Product name	U10	U10	U10	RI70	RI60	RI60	RI60
No-load speed [deg/s]	17,189	17,189	17,189	18,750	20,970	20,970	20,970
Max torque [Nm]	5.04	5.04	5.04	2.68	1.63	1.63	1.63
Power coefficient	124.57	124.57	124.57	33.44	3.54	3.54	3.54
Motor controller efficiency (output/regeneration)	0.9 / 0.8	0.9 / 0.8	0.9 / 0.8	0.9 / 0.8	0.9 / 0.8	0.9 / 0.8	0.9 / 0.8

*Reduction ratios between motors and joints at a standing pose.

regeneration efficiency of the motor controller, respectively. Unlike human muscles, motors are capable of both dissipating and recovering energy. A portion of the output energy is lost due to inelastic collisions between the foot and the ground, as well as friction, sound, and heat, allowing only partial energy regeneration. By minimizing energy loss during impact with the ground, P_{mech} can be reduced. This characteristic differs from biological systems, where muscles cannot recover energy, but tendons store and release energy through their elasticity. However, the energy output efficiency and regeneration efficiency vary depending on the configuration of the motor controller. In the case of the motor controller we developed, the estimated output efficiency and regeneration efficiency are $\eta_{fwd} = 0.9$ and $\eta_{inv} = 0.8$, respectively.

Fig. 4 (f) and (g) show 4 contact points at the toe and 1 contact point at the heel. In real-world scenarios, stable surface contact between the foot and the ground enhances friction and contributes to stable locomotion. However, replicating this effect accurately in simulation is challenging. Therefore, instead of modifying the contact model, we implemented a reward function design that encourages at least three-point contact to promote stable interaction with the ground.

B. Reinforcement Learning Setup

We defined our task as a Markov Decision Process (MDP) and employed Reinforcement Learning. Considering deployment in the real world, similar to the research conducted by Qiayuan Liao et al. [5], we primarily utilized proprioceptive data that can be obtained with relatively high accuracy in the real world. The observation space comprises the base linear velocity [24] (3 elements), base angular velocity (3 elements), projected gravity vector (3 elements), current command vector (3 elements), joint pose and joint

velocity, as well as the previous action (with 12 elements for toe lock and 14 elements for toe unlock). The action space is defined as the joint positions, consisting of 12 dimensions for the robot without toes and 14 dimensions for the robot with toes. These actions are used for the PD controllers, where the target joint torques are converted to the motor torques, and after applying actual constraint, then reconverted into joint torques to be used for the simulator.

For policy optimization, we employed a Multi-layer Perceptron (MLP) with hidden layers of sizes [512, 256, 128] and an ELU activation function, utilizing Proximal Policy Optimization (PPO) [25] as implemented in the ISAAC Lab [26] environment. While a uniform noise is added to the PPO actor, the critic was implemented without noise, forming an asymmetric actor-critic architecture. The policy is computed at 50 Hz, with a simulation step time of 0.005 seconds.

To compare efficiency and ground impact properties, a normal walking scenario at 1.3 m/s was constructed. Additionally, to examine the influence of toe joints on agile motions, a new test benchmark, inspired by a standard benchmark test for human athletes, was proposed and applied.

IV. VALIDATION

A. Efficient Walking and Impact Absorption

To compare the efficiency and impact absorption properties of bipedal robots with and without toes, a validation study using a walking simulation at 1.3 m/s, which is equivalent to the human's normal walking speed, was conducted. Three distinct cases were established: (i) a robot with toes, (ii) a robot without toes, but with the sole constrained to remain horizontal to the ground, and (iii) a robot without toes, but allowing heel-to-toe motions. Gait patterns such as (ii) are commonly used in many bipedal robots to ensure stable ground contact. However, (iii) is considered challenging because contact with the edge of the sole can be unstable.

The reward functions for RL were set similarly to [5], and for an objective comparison, subjective rewards and penalties—such as penalties for vertical (Z-directional) movements and joint deviations that encourage a predefined default pose—were either removed or reduced using curriculum learning. However, the following reward functions were utilized to ensure accurate and clear comparisons:

- An accurate CoT value, computed using Equation (6), was applied as a penalty function.
- To encourage stable surface contact, a reward function was applied when 3 or more contact points were engaged.
- For the case (ii), a reward function was introduced to directly promote maintaining the sole in a horizontal orientation relative to the ground.

The three cases were compared using a baseline model that converged to a sufficiently low CoT value and tracked the commanded velocity with an error margin within 3%.

Fig. 5 (a), (e), and (i) show snapshots of walking for the three cases. As observed from the contact points indicated by red circles in the figures, cases (i) and (ii) maintained surface contact for a longer period than case (iii). Fig. 5 (b), (f), and (j) depict the torque of each joint, while Fig. 5 (c), (g), and (k)

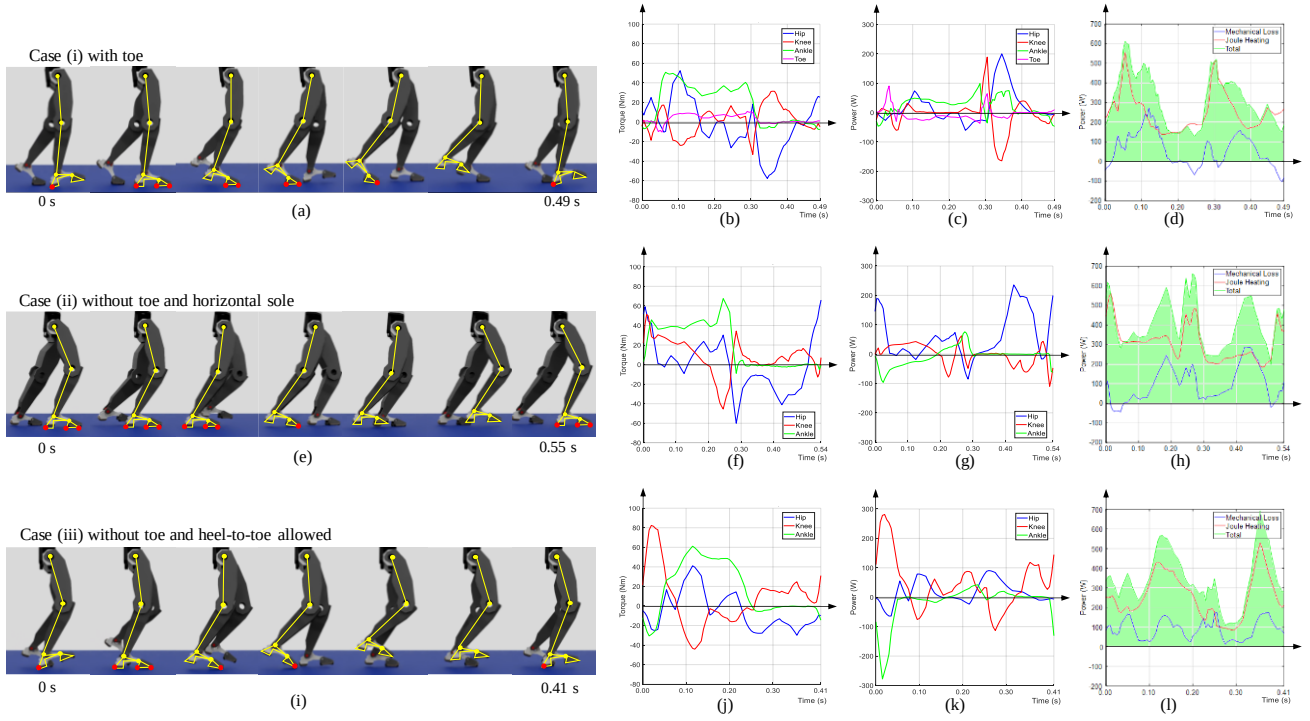


Fig. 5 Simulation results of 1.3 m/s walking. (a), (b), (c), and (d) show the results of case (i) with toes. (e), (f), (g), and (h) present the results of case (ii) without toes, maintaining the sole horizontal to the ground. (i), (j), (k), and (l) display the result of case (iii) walking without toes, allowing heel-to-toe motions. (a), (e), and (i) show snapshots of the right leg, with contact points marked by red circles. (b), (f), and (j) illustrate joint torques. (c), (g), and (k) depict the power consumption of each joint. (d), (h), and (l) present the total power consumption, including mechanical loss and Joule heating loss.

Table II. Power and GRF comparison of 1.3 m/s walking

Items	Case (i) with toes	Case (ii) without toes & horizontal sole	Case (ii) without toes & heel-to-toe
Total Power [W]	384.27	430.32	414.90
Mechanical loss [W]	124.03	141.38	148.64
Joule heating [W]	257.24	288.94	266.22
COT	0.934	1.054	1.017
Max GRF [N]	770.58	1165.24	2089.67

show the power consumption of each joint. Fig. 5 (d), (h), and (l) illustrate the total power consumption, which consists of Joule heating and mechanical losses. Case (ii) exhibited a large Joule heating loss because the knee was bent, requiring higher torques than the other cases. In contrast, case (i) showed low Joule heating loss as well as minimal mechanical loss, as the knee remained near a straight posture, and ground contact losses were reduced. Table II summarizes the results. An interesting observation is that even in the absence of toes, adopting a human-like heel contact to toe-off motion as seen in Case (iii) improved energy efficiency. Case (iii) showed the result similar to (i) except the maximum ground reaction force (GRF). As inferred from the values of GRF, the robot with toes can significantly reduce the impact from the ground.

B. Agility Test

To investigate the effect of toes on the agility of bipedal robots, a benchmark test called the Robot T-Test was proposed, inspired by the standard agility benchmark for athletes, the Agility T-Test [28]. To focus on testing agility rather than sprinting, the size of the test set was reduced, and turning orientation was allowed, as shown in Fig. 6(a). The reward functions were similar to those used for 1.3 m/s walking in the previous subsection. However, the CoT penalty and rewards for periodic walking were significantly reduced. To train the generalized capability for agile motions rather than merely learning the specific Robot T-Test task, the command was set as follows: The linear velocities in the x and y directions and the angular velocity in the z direction were arbitrarily selected within the ranges of (-2.0 to 4.0) m/s, (-2.0 to 2.0) m/s, and (-6.3 to 6.3) rad/s, respectively. The commands were updated within 5 to 7 seconds.

Fig. 6(b) and (c) show snapshots of the turning motions for the robots with and without toes, respectively. As observed, the robot with toes actively utilized the toe joints, while the robot without toes maintained a lower pelvis height to compensate for the lack of toe joints and achieve agile motion. Fig. 7(a) and 7(b) show the robot trajectories during the T-test. As seen in the figures, the trajectory of the robot with toes closely follows the intended T-shape, while the robot without toes deviates significantly from the prescribed trajectory at the turning points. In addition, Fig. 7(c) shows that the model without toes has a lower vertical (Z-axis) position compared

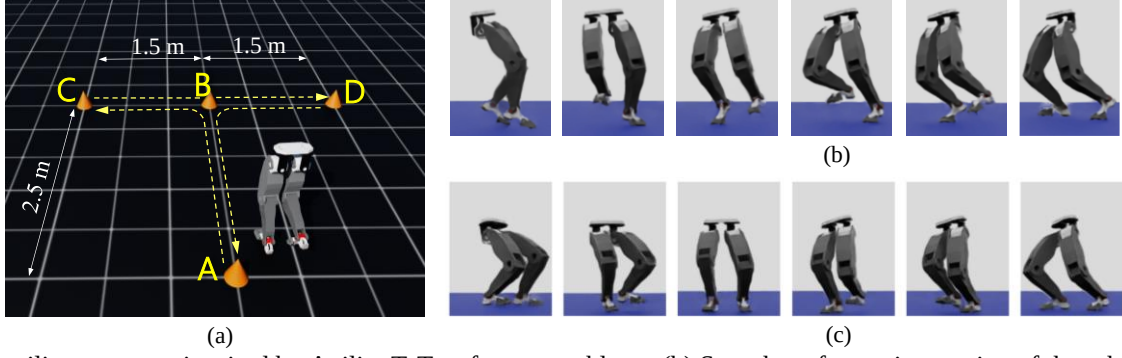


Fig. 6 (a) agility test setup inspired by Agility T-Test for sport athletes. (b) Snapshot of a turning motion of the robot with toes near the point C. (c) Snapshot of a turning motion of the robot without toes near the point C.

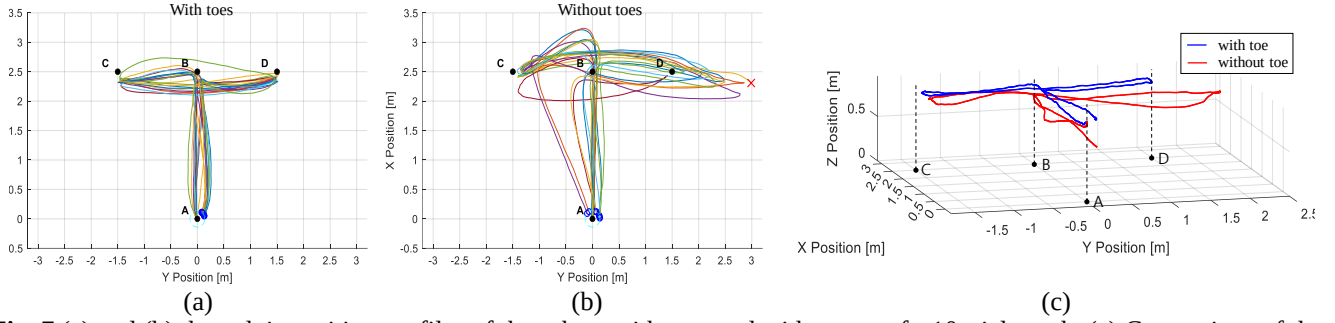


Fig. 7 (a) and (b) the pelvis position profiles of the robots with toes and without toes for 10 trials each. (c) Comparison of the profiles at 3D space, where blue curve and red curves indicate the robot with toes and without toes respectively.

to the model with toes, which suggests a deliberate lowering of the center of mass to enhance stability and prevent falling during trajectory tracking. As shown in Table III, the completion time for the robot with toes was on average 1.21 seconds shorter, the mean deviation from the T-test trajectory was reduced by 0.117 m, and the GRF was lowered by 482.6 [N]. Conversely, the average speed was higher for the model without toes, as the model with toes decelerated more effectively during the turning segments.

Consequently, the simulation results show that active toes contribute to enhanced robotic agility. The presence of toes enables the robot to extend its foot further during directional changes, generating a larger rotational moment. In other words, the model with toes, which maintains stable ground contact over a wider range of motion, demonstrates greater robustness in executing directional changes.

C. Limitations and Discussion

Despite developing a rigorous simulation model from the component level to objectively compare and analyze the expected characteristics and performance of the robot, several limitations are remained. The inertia of motors with coupled mechanisms appears as an off-diagonal term in the inertia matrix of the dynamic equation, but the Isaac Sim simulation algorithm could not incorporate this effect. Instead, the inertia applied to the joints was conservatively set to a higher value. In highly dynamic motions, a broad and stable ground contact can provide benefits in real-world scenarios, but accurately modeling this effect in simulation is challenging. To address this, the reward function was designed to encourage stable

Table III. Agility Test Results

Items	With Toes	Without Toes
No. of success /No. of trial	10/10	9/10
Average completion time	9.03 s	10.24 s
Average speed	1.16 m/s	1.30 m/s
Position error from the path	0.139 m	0.256 m
Average of peak GRF for each step	1918.1 N	2400.7 N

surface contact, though this approach may influence fair comparisons. Further validation is needed for sprinting, jumping, ascending, and walking on rough terrains. Curriculum learning was employed to refine the training process, ultimately reducing developer intervention by leaving only minimal high-level objective rewards. Nonetheless, a broader range of objective comparisons is required to accurately evaluate the characteristics, advantages, and limitations of the proposed toe mechanism and CA structure.

The results of this study are not generally applicable to all robots with toes, but limited to the proposed structure featuring compliant actuation (CA) and an active toe mechanism. Additionally, energy regeneration in robots is a characteristic that does not exist in humans, while the efficient eccentric motion of human muscles and the elasticity of tendons, which enable energy storage and release, are not

implemented in robots. Therefore, direct comparisons of gait and CoT between the human and the robot based on the results have limitations.

V. CONCLUSION

This paper proposes a 14-DOF bipedal robot equipped with active toes and evaluates its walking efficiency, impact absorption, and agility through RL-based simulation. The knee joint, two-DOF ankle joint, and toe joint are intentionally coupled to generate high forces during walking, while all heavy actuators are placed in the thigh, resulting in a lightweight and robust lower leg structure. To minimize the sim-to-real gap, the simulation was set to accurately reflect the coupled characteristics of the proposed robot, ensures realistic ground contact for stable walking, and incorporates precise CoT calculations. The training process was designed with an objective reward function to minimize developer bias. As a result, the simulation results indicate that the bipedal robot with active toes demonstrated a 11.3% reduction in CoT, a 33.8% reduction in maximum GRF at a walking speed of 1.3 m/s, and a 10.2% reduction in completion time for agility motions compared to the same robot without toe joints.

As future work, we plan to refine the design based on simulation results and proceed with building and experimentally validating the real robot. We hope that this study contributes to the advancement of robots toward a future where they can operate safely, quietly, and dynamically alongside humans.

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